

MaskAgent: A Privacy-First On-Device AI Browser Agent for Secure Web Automation

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Abstract

The rapid advancement of Large Language Models (LLMs) and Vision-Language Models (VLMs) has enabled intelligent browser agents capable of understanding webpages and performing automated tasks such as clicking, typing, scrolling, and navigation. However, conventional AI browser agents may expose webpage content, screenshots, and user-interface information to external AI services during reasoning, creating privacy concerns when webpages contain personally identifiable information (PII), financial data, healthcare information, authentication data, or confidential organizational content. This paper presents **MaskAgent**, a privacy-first on-device AI browser agent that introduces a local privacy protection layer between the user's browser and the AI reasoning system. The proposed architecture follows the principle **“Perceive Locally, Protect Locally, Reason Intelligently.”** MaskAgent combines DOM-based analysis and visual perception to understand webpage content locally, detects sensitive information, masks or redacts sensitive regions, and generates sanitized context before AI reasoning. The sanitized context is then processed by an LLM or VLM to determine the required browser action, which is subsequently executed through the browser extension. The system uses browser-extension technologies, JavaScript/TypeScript, WebAssembly, WebGPU, lightweight vision models, ONNX Runtime Web, and Transformers.js. MaskAgent is intended for privacy-sensitive applications including banking, healthcare, education, government, and enterprise environments. The proposed evaluation uses visual accuracy, PII precision and recall, redaction precision, latency, and resource utilization.

Index Terms— Artificial intelligence, browser automation, on-device artificial intelligence, personally identifiable information, privacy-preserving artificial intelligence, redaction, web agents.

I. INTRODUCTION

Artificial intelligence has evolved from systems that provide text-based responses into autonomous agents capable of interacting with digital environments. Modern web agents can interpret natural-language instructions and perform browser operations such as clicking buttons, entering information, scrolling webpages, and navigating between pages. These capabilities have created opportunities for intelligent automation in everyday web applications.

Recent research has demonstrated the ability of language and multimodal models to operate websites and perform complex tasks. Mind2Web introduced a large-scale dataset for developing generalist web agents and demonstrated the challenges associated with interacting with real-world websites [1]. WebArena provided a realistic environment for evaluating autonomous web agents on functional websites [2]. WebVoyager demonstrated an end-to-end web agent using large multimodal models to interact with real websites through visual and textual information [3].

Although these systems demonstrate significant progress, privacy remains an important challenge. A webpage can contain names, email addresses, telephone numbers, addresses, financial information, medical information, account details, and other confidential information. If raw webpage content or screenshots are provided to a remote AI model, sensitive information may be exposed outside the user's device.

The central challenge is therefore to provide intelligent browser automation without requiring unrestricted access to raw sensitive screen content.

MaskAgent addresses this challenge through a privacy-first architecture. Instead of directly transmitting webpage information to an external AI model, MaskAgent performs perception and sensitive-information protection locally. The system identifies webpage elements, detects potentially sensitive information, masks or redacts sensitive regions, and creates a sanitized representation. Only the sanitized context is supplied to the AI reasoning component.

The proposed project follows the principle:

Perceive Locally → Protect Locally → Reason Intelligently

The main contributions of MaskAgent are:

1. A privacy-first architecture for AI-powered browser automation.
2. Local DOM and visual webpage perception.
3. Local detection and redaction of PII and sensitive information.
4. A sanitized-context privacy boundary before AI reasoning.
5. Integration of AI reasoning with browser automation.
6. An evaluation framework based on perception accuracy, privacy detection, redaction, latency, and resource usage.

II. RELATED WORK

A. Web-Based AI Agents

Web agents combine natural-language reasoning with browser interaction. Mind2Web demonstrated that web agents can be trained to perform tasks across a wide range of real websites [1]. The work highlights the difficulty of generalizing web interaction because websites differ significantly in structure, layout, and interaction mechanisms.

WebArena introduced a realistic evaluation environment containing functional websites and tasks designed to evaluate autonomous web agents [2]. Such environments provide a basis for studying the reliability and generalization of browser-based agents.

MaskAgent addresses an additional dimension of browser agents: privacy. Rather than focusing only on task completion, the proposed system introduces local sensitive-information protection before AI reasoning.

B. Vision-Language Web Agents

Visual understanding is increasingly important for browser automation because webpages contain information that may not be completely represented through DOM structures. WebVoyager demonstrated the use of a multimodal model for interacting with real-world websites using visual and textual observations [3].

MaskAgent similarly uses visual perception but places privacy protection before external reasoning. Sensitive visual regions are detected and masked locally so that the reasoning model receives sanitized information.

C. On-Device Machine Learning

On-device machine learning enables inference to be performed closer to the data source. Browser-compatible frameworks such as ONNX Runtime Web provide mechanisms for running machine-learning models using JavaScript and browser execution backends [4], [5]. Transformers.js provides browser-based transformer inference and supports WebGPU-based acceleration for compatible models [6], [7].

III. PROBLEM STATEMENT

Conventional AI browser agents require webpage information to determine which action should be performed. This information may contain sensitive user or organizational data.

A conventional architecture can be represented as:

Browser → Raw Webpage/Screen Data → External AI → Browser Action

In this model, the AI system may receive raw information from the user's browser.

The problem addressed by MaskAgent is:

How can an AI browser agent perform intelligent web automation while minimizing exposure of sensitive webpage and screen information to an external AI reasoning system?

The proposed solution is:

Browser → Local Perception → Sensitive Data Detection → Redaction → Sanitized Context → AI Reasoning → Browser Action

The objective is to preserve sufficient contextual information for intelligent reasoning while preventing unnecessary exposure of sensitive information.

IV. PROPOSED SYSTEM

A. Overview

MaskAgent is a privacy-first browser agent that combines local webpage perception, sensitive-information detection, privacy masking, AI reasoning, and browser automation.

The core processing pipeline is:

Browser Screen → DOM + Visual Perception → PII Detection and Redaction → Sanitized Context → LLM/VLM Reasoning → Browser Action

The system consists of the following major components:

1. Browser Extension
2. DOM Analysis Module
3. Visual Perception Module
4. PII Detection Module
5. Privacy Masking Module
6. Sanitized Context Generator
7. LLM/VLM Reasoning Module
8. Browser Action Executor

B. Browser Extension

The browser extension acts as the primary interface between the user and MaskAgent. It monitors the webpage, collects relevant webpage information, processes the information locally, and executes browser actions generated by the reasoning module.

The prototype is based on a Manifest V3 extension architecture [9]. The extension supports webpage interaction, DOM access, content analysis, local processing, browser action execution, and communication with the AI reasoning component.

V. SYSTEM ARCHITECTURE

The architecture of MaskAgent can be divided into two logical environments.

A. Local Privacy Environment

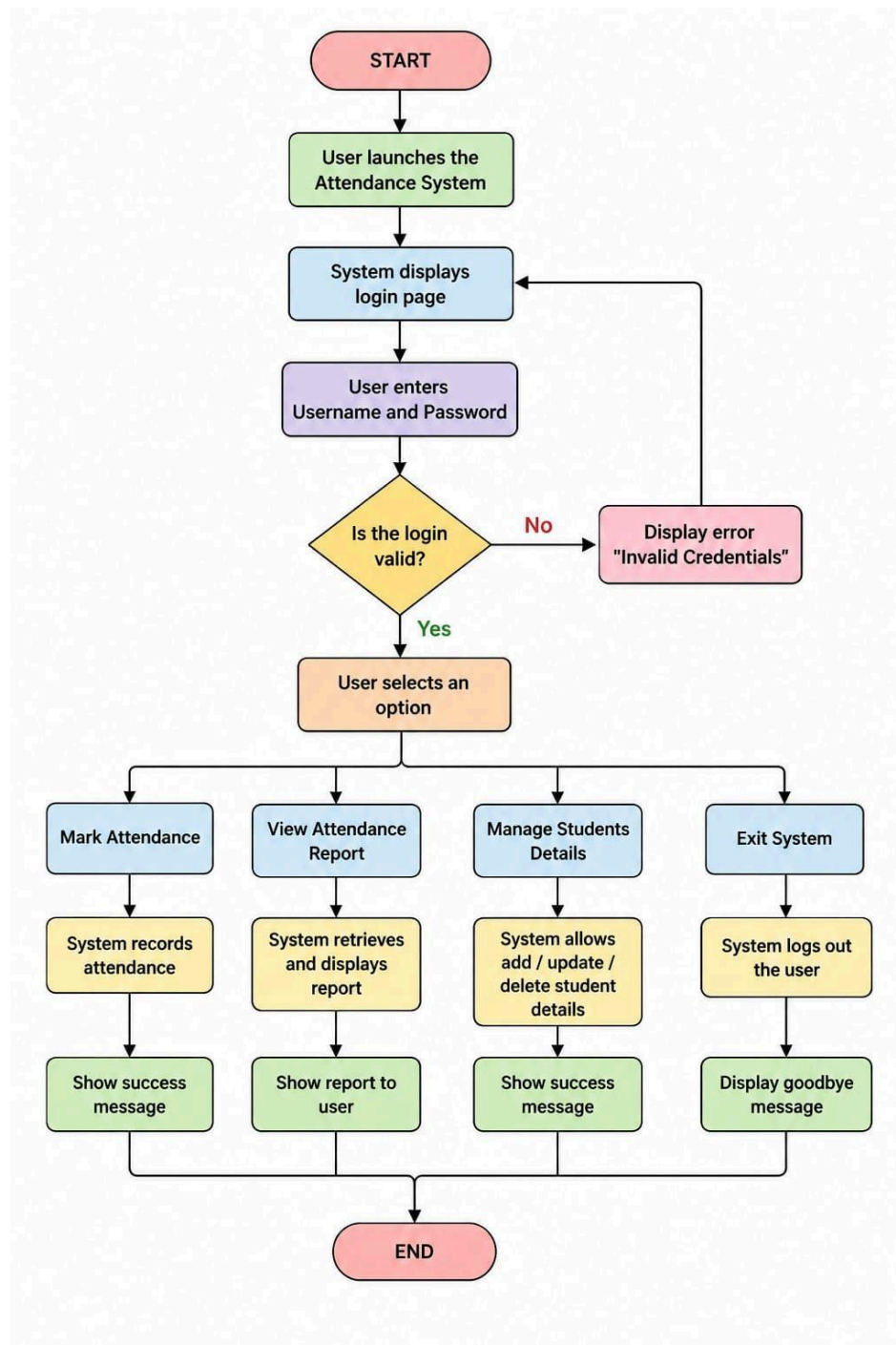
The local environment contains the browser, DOM analyzer, visual perception model, PII detector, redaction module, and sanitized-context generator.

B. AI Reasoning Environment

The reasoning environment contains the LLM/VLM, task interpretation, and action-selection components.

The privacy boundary is placed between the sanitized-context generator and the AI reasoning component.

Fig. 1. Proposed MaskAgent Architecture



The key feature of the architecture is that raw sensitive screen information is processed before it reaches the AI reasoning stage.

VI. LOCAL WEBPAGE PERCEPTION

A. DOM-Based Perception

The Document Object Model (DOM) represents the structure of a webpage and provides information about elements such as buttons, links, forms, text fields, labels, images, and navigation components.

MaskAgent uses DOM information to identify interactive elements, text content, element hierarchy, input fields, buttons, links, labels, and element positions.

B. Visual Perception

DOM information alone may not completely represent the visual state of a webpage. Modern webpages can contain dynamically rendered elements, canvas content, images, and visually arranged components.

Therefore, MaskAgent incorporates visual perception to identify text regions, buttons, icons, input fields, layout structures, sensitive visual regions, and interactive elements.

C. DOM and Vision Fusion

The combination of DOM and visual information provides complementary information. DOM analysis provides structural and semantic information, whereas visual perception provides information about the rendered appearance of the page.

The proposed system uses DOM and vision fusion to improve webpage understanding and sensitive-content detection.

VII. PII AND SENSITIVE INFORMATION DETECTION

The PII detection module identifies information that should not be directly exposed to the AI reasoning component.

Potential categories include names, email addresses, telephone numbers, physical addresses, identification numbers, financial information, medical information, account information, authentication-related information, and confidential organizational information.

For a detected region i , a confidence score can be represented as:

$$C_i = P(S_i|X)$$

where S_i represents a sensitive-information category and X represents the observed webpage content.

A detection threshold can be defined as:

$$D(X) = \{1, C_i \geq T; 0, C_i < T\}$$

where T represents the selected detection threshold.

The actual threshold should be determined experimentally based on the implemented detection model.

VIII. PRIVACY MASKING AND REDACTION

After sensitive information has been detected, MaskAgent performs local masking or redaction. The purpose of redaction is to remove sensitive values while retaining enough contextual information for AI reasoning.

For example:

Original Webpage Content

Name: John Smith

Email: johnsmith@example.com

Phone: 9876543210

Sanitized Representation

Name: [PERSON]

Email: [EMAIL]

Phone: [PHONE]

The semantic meaning of the webpage is preserved while the original sensitive values are replaced.

For visual data, sensitive areas can be represented using bounding boxes:

$$B_i = (x_i, y_i, w_i, h_i)$$

where x_i and y_i indicate the position of the region, while w_i and h_i represent its dimensions.

The masking operation can be represented as:

$$I'(y) = \{M, (y) \in B_i I(y), \text{ otherwise}$$

where I is the original image, I' is the sanitized image, and M is the masking representation.

IX. SANITIZED CONTEXT GENERATION

Sanitized context generation forms the main privacy boundary of MaskAgent.

Let:

- C_r = raw webpage context
- R = detected sensitive regions

- C_s = sanitized context

Then:

$$C_s = f(R)$$

The function f replaces or removes sensitive information while preserving task-relevant context.

For example, a banking webpage may contain account information, transaction details, and navigation buttons. MaskAgent can retain the navigation structure and interaction elements while replacing account-specific values with privacy-safe placeholders.

The central principle is:

Raw sensitive data stays local; only sanitized context reaches AI.

X. AI REASONING MODULE

The sanitized webpage context is provided to an LLM or VLM.

The reasoning module receives:

1. User instruction.
2. Sanitized webpage context.
3. Available browser actions.

The reasoning process can be represented as:

$$Instruction + C_s \rightarrow AI\ Reasoning \rightarrow Action$$

The AI determines the appropriate action based on the available sanitized information.

Supported browser actions include click, type, scroll, and navigate.

For example:

User Instruction: “Open the settings page.”

Sanitized Webpage Context:

Navigation:

Home | Profile | Settings | Help

AI Output:

CLICK(Settings)

The generated action is then passed to the browser execution module.

XI. BROWSER ACTION EXECUTION

The action execution module converts the AI-generated action into a browser operation.

$Action \in \{ckTypeScrollNavigate\}$

A complete agent cycle can therefore be represented as:

$Observe \rightarrow Detect \rightarrow Sanitize \rightarrow Reason \rightarrow Act$

After execution, the browser can be observed again to determine the next state, creating an iterative interaction loop.

XII. IMPLEMENTATION TECHNOLOGIES

A. JavaScript and TypeScript

JavaScript and TypeScript can be used for browser-extension development, DOM processing, browser interaction, communication between modules, and user-interface development.

B. WebAssembly

WebAssembly can provide efficient execution for suitable machine-learning workloads inside browser environments.

C. WebGPU

WebGPU can be used for GPU-accelerated local AI inference on supported devices [8].

D. ONNX Runtime Web

ONNX Runtime Web provides browser-based machine-learning inference using JavaScript and supports execution backends such as WebAssembly and WebGPU [4], [5].

E. Transformers.js

Transformers.js enables transformer models to run directly in browser environments and supports WebGPU acceleration for compatible models [6], [7].

F. Lightweight Vision Models

Lightweight models are preferred for local perception because browser environments have limited computational resources compared with dedicated AI servers.

XIII. PRIVACY-PRESERVING ARCHITECTURE

The main privacy property of MaskAgent is that sensitive information is processed before AI reasoning.

Let D represent raw webpage data and S represent sensitive information contained within D :

$$S \subset D$$

After sanitization, the resulting data D' should minimize the presence of S :

$$S \not\subset D'$$

At the same time, sanitized data should preserve task-relevant information:

$$Utility(D') \approx Utility(D)$$

Therefore, MaskAgent attempts to optimize two objectives:

1. Privacy preservation.
2. Task usefulness.

The system acts as a local privacy firewall between the user's browser and the AI reasoning component.

XIV. FEASIBILITY AND VIABILITY

MaskAgent is based on an existing prototype and is proposed as an enhancement toward an SIH-ready privacy-preserving browser agent [9].

The existing prototype capabilities include:

- Manifest V3 extension.
- DOM/UI analysis.
- PII detection and redaction.
- Local AI integration.
- Browser automation.

A. Computational Resources

Local AI inference can require significant computational resources. Lightweight and optimized models can improve browser compatibility.

B. Inference Latency

WebGPU acceleration can reduce inference time on supported devices.

C. Detection Accuracy

Combining DOM and visual perception can improve detection of sensitive content and webpage elements.

D. Redaction Precision

Precise region detection can reduce unnecessary masking and preserve useful context.

E. Privacy Protection

Privacy-preserving processing is performed before sensitive information is sent beyond the local privacy boundary.

XV. EXPERIMENTAL METHODOLOGY

The proposed evaluation uses synthetic webpages containing both ordinary and sensitive information. Real personal information should not be used for testing.

A. Test Environment

The system can be evaluated using Chrome or Firefox browsers, different webpage layouts, different screen resolutions, synthetic PII datasets, different device configurations, and different AI model configurations.

B. Test Cases

The evaluation can include:

1. Plain-text PII.
2. PII inside forms.
3. PII inside tables.
4. PII inside images.
5. Financial information.
6. Healthcare-style information.
7. Login-style interfaces.
8. Dynamic webpages.
9. Multiple sensitive regions.
10. Pages containing both sensitive and non-sensitive information.

C. Browser Automation Tasks

. The following tasks can be evaluated:

- Click a specified button.
- Enter non-sensitive information.
- Navigate to a webpage.
- Scroll to a particular section.
- Select a UI element.
- Perform a multi-step navigation task.

XVI. PERFORMANCE EVALUATION

The proposed evaluation uses five primary categories:

1. Visual Accuracy.
2. PII Precision and Recall.
3. Redaction Precision.
4. Latency.
5. Resource Usage.

A. PII Detection Precision

$$Precision = \frac{TP}{TP+FP}$$

where TP represents correctly detected sensitive regions and FP represents incorrectly detected regions.

B. PII Detection Recall

$$Recall = \frac{TP}{TP+FN}$$

where FN represents sensitive regions that were not detected.

C. F1 Score

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

D. End-to-End Latency

$$T_{total} = T_p + T_d + T_r + T_s + T_a$$

where T_p is perception time, T_d is detection time, T_r is redaction time, T_s is AI reasoning time, and T_a is browser action time.

E. Resource Usage

Resource consumption can be evaluated using CPU utilization, GPU utilization, RAM consumption, model size, and network data transmitted.

TABLE I
PROPOSED EVALUATION METRICS

Metric	Measurement
Visual Accuracy	Correct webpage/UI element identification

Metric	Measurement
PII Precision	Correct sensitive-region detections
PII Recall	Sensitive regions successfully detected
Redaction Precision	Correctly masked sensitive regions
Latency	End-to-end task completion time
Resource Usage	CPU, GPU, memory, and network usage

XVII. RESULTS AND DISCUSSION

The final experimental results should be presented using actual measurements from the MaskAgent implementation.

TABLE II
EXPERIMENTAL RESULTS

Evaluation Metric	MaskAgent Result
Visual Accuracy	[Insert %]
PII Precision	[Insert %]
PII Recall	[Insert %]
F1 Score	[Insert %]
Redaction Precision	[Insert %]
Average Latency	[Insert ms]
Memory Usage	[Insert MB]
Network Data Exposure	[Insert measurement]

Unsupported performance values should not be reported. The results should be analyzed according to the relationship between privacy protection and browser-agent functionality.

A successful implementation should demonstrate that sensitive information can be detected and redacted locally while sufficient webpage context remains available for AI reasoning and browser automation.

The evaluation can also compare DOM-only perception, visual-only perception, DOM and visual perception, CPU-based inference, and WebGPU-based inference.

XVIII. APPLICATIONS

A. Banking

Banking websites contain sensitive account information and financial data. MaskAgent can assist with navigation and interface interaction while reducing unnecessary exposure of sensitive information.

B. Healthcare

Healthcare portals can contain sensitive patient information. Local detection and redaction can reduce unnecessary exposure during AI-assisted interaction.

C. Education

Educational portals may contain student information and academic records. Privacy-preserving browser automation can support intelligent interaction with such systems.

D. Government

Government websites can contain personally identifiable and confidential information. MaskAgent can provide AI-assisted browser navigation while protecting sensitive content.

E. Enterprise

Enterprise dashboards can contain confidential organizational data. MaskAgent can assist users with browser-based workflows while minimizing exposure of sensitive information.

XIX. IMPACT AND BENEFITS

A. Privacy

MaskAgent protects sensitive visual and textual information by detecting and masking it before AI reasoning.

B. Intelligent Automation

The system enables AI-assisted browser workflows involving clicking, typing, scrolling, and navigation.

C. On-Device AI

Local perception reduces the need to transfer raw webpage information to external services.

D. Reduced Data Exposure

Only sanitized context is intended to cross the privacy boundary.

E. Practical Deployment

Browser-extension technology allows the system to operate within commonly used web browsers.

The intended impact can be summarized as:

Privacy Without Losing AI Capability.

XX. FUTURE WORK

Future development of MaskAgent can focus on improving the accuracy, efficiency, and adaptability of the proposed architecture. Potential areas include:

1. Improved multimodal PII detection.
2. More efficient lightweight vision models.
3. Quantized local models for low-resource devices.
4. Improved DOM-vision fusion.
5. Adaptive privacy policies.
6. Fine-grained region-based masking.
7. Faster WebGPU inference.
8. Fully offline AI reasoning.
9. Risk-aware browser action validation.
10. User-configurable privacy levels.
11. Larger-scale benchmarking.
12. Improved support for dynamically generated webpages.

XXI. CONCLUSION

This paper presented MaskAgent, a privacy-first on-device AI browser agent designed to combine intelligent web automation with local sensitive-information protection. The proposed architecture addresses the privacy challenge associated with AI browser agents by placing a local perception and redaction layer between the user's browser and the AI reasoning component.

The system follows the principle:

Perceive Locally → Protect Locally → Reason Intelligently.

MaskAgent analyzes webpages using DOM and visual perception, detects PII and sensitive information locally, masks or redacts sensitive regions, generates sanitized context, and provides that context to an LLM or VLM for reasoning. The resulting action is then executed through the browser extension.

The architecture uses Manifest V3, JavaScript/TypeScript, WebAssembly, WebGPU, lightweight vision models, ONNX Runtime Web, and Transformers.js. The design is intended to support privacy-sensitive applications including banking, healthcare, education, government, and enterprise environments.

By placing a privacy boundary before AI reasoning, MaskAgent aims to demonstrate that useful browser automation can be achieved without requiring unrestricted exposure of raw sensitive screen content.

Protect First. Then Reason.

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